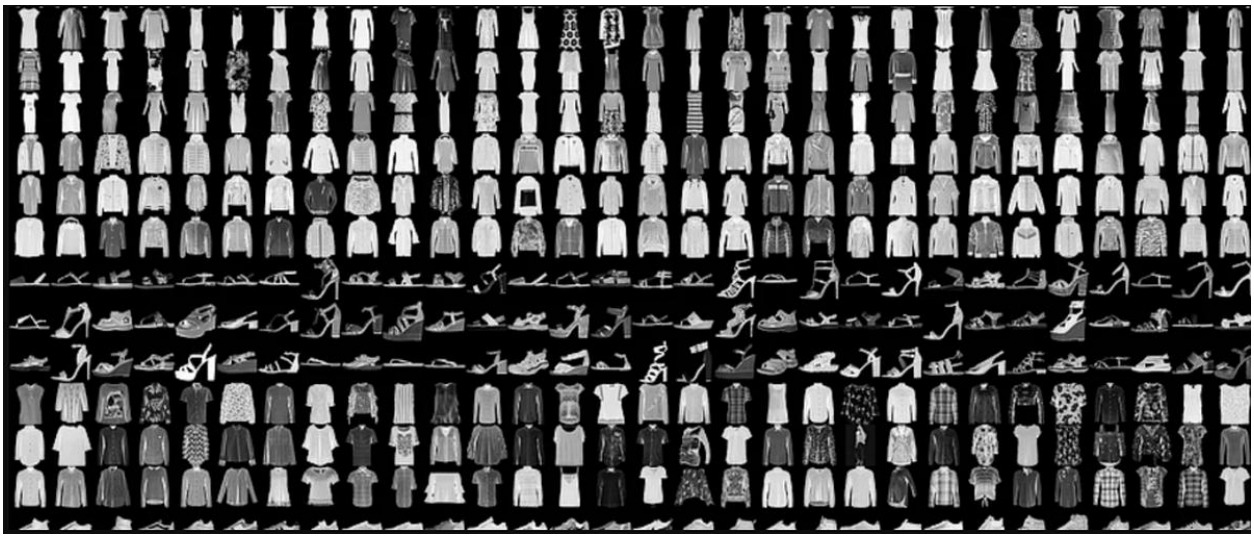
Lab 1 – MLP Autoencoder

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Part 1: Prelab



Step 1 – IDE Installation

There are a few main integrated development environment (IDE) options for this course.

- One IDE option is PyCharm.
  - https://www.jetbrains.com/pycharm/download/
- Another option is Visual Studio Code (VS Code).
  - https://code.visualstudio.com/download?_exp_download=d53503e735
- Google Colab is an option that is available on the web.
  - https://colab.research.google.com/

Step 2 – Project Setup

For Colab:

1. Create a folder within your Google Drive. All code will be placed within this folder.
2. No venv setup required. Instead, the drive must be mounted to the notebook. Steps are given here: https://colab.research.google.com/notebooks/io.ipynb

For WSL:

1. Create a local project folder for this lab.
2. From within the WSL shell navigate to the local project folder:

```
cd path/to/your/project
```

3. Create the virtual environment (e.g. elec475_venv):

```
conda create --name elec475_env
```

4. Activate the virtual environment:

```
Conda activate elec475_venv
```

For other: Setup in Terminal / Command Prompt:

1. Create a project folder for this lab.
2. Open Terminal / Command Prompt.
   a. Navigate to the project directory:

```
cd path/to/your/project
```

b. Create the venv:
    i. Windows:

```
python -m venv .venv
```

    ii. MacOS / Linux:

```
python3 -m venv .venv
```

c. Activate the venv:

    i. Windows:

```
.venv\Scripts\activate.bat
```

    ii. MacOS / Linux:

```
source .venv/bin/activate
```

Once the virtual environment has been activated, the required libraries can be installed as follows:

```
pip install torch torchvision matplotlib numpy torchinfo
```

Step 3 – FashionMNIST Download

The dataset being used for this lab will be FashionMNIST. This dataset should be downloaded and available to you before the beginning of the lab. This script should download the dataset for you:

```
from torchvision.datasets import FashionMNIST
import torchvision.transforms as transforms

DATA_DIR = './data'
transform = transforms.Compose([transforms.ToTensor()])
train_set = FashionMNIST(DATA_DIR, train=True, download=True, transform=transform)
test_set = FashionMNIST(DATA_DIR, train=False, download=True, transform=transform)
```

```
print('train:', len(train_set), 'test:', len(test_set))
img, label = train_set[0]
print('shape:', img.shape, 'dtype:', img.dtype)
print('min:', img.min().item(), 'max:', img.max().item())
print('raw dtype:', train_set.data.dtype, 'raw max:', train_set.data.max().item())
```

If using Colab, **DATA_DIR** must be replaced with a location within your Google Drive. For example: **'/content/drive/MyDrive/elec475/lab1/data'**

Step 4 – FashionMNIST Visualization

The final step of the prelab is to visualize the images within your IDE:

```
import matplotlib.pyplot as plt
import torchvision.transforms as transforms
from torchvision.datasets import FashionMNIST

DATA_DIR = './data'

CLASSES = ['T-shirt/top', 'Trouser', 'Pullover', 'Dress', 'Coat',
           'Sandal', 'Shirt', 'Sneaker', 'Bag', 'Ankle boot']

train_set = FashionMNIST(DATA_DIR, train=True, download=False,
                         transform=transforms.Compose([transforms.ToTensor()]))

idx = int(input('index > '))
img = train_set.data[idx]
label = train_set.targets[idx].item()

print('label =', label, '(' + CLASSES[label] + ')')
plt.imshow(img, cmap='gray')
plt.title(CLASSES[label])
plt.show()
```